

User Control in Tourism Recommender Systems

Group W3

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TUM Uhrenturm

Project Overview & Motivation

- **The Recommender “Black-Box” Problem:** Traditional recommender systems hide their logic, leading users left unknown. [1].
- **Core Goal:** Give users information and control over the ranking logic and filtering mechanisms [4].
- **Regulatory Motivation: EU Digital Services Act (DSA):**
 - *Article 27 (Recommender System Transparency):* Online platforms must explain the main parameters used in recommender systems and provide user-adjustable options [2].
 - *Article 38 (Profiling Options):* Users must be offered at least one option that is not based on profiling [2].

System Architecture & Tech Stack

Fully implemented a working responsive web application:

- **Frontend Framework:** React 18¹ + Vite² for rapid, lightweight development.
- **Mathematical Formula Rendering:** Uses KaTeX³ to render equations directly in browser.
- **Dataset:** Mock database of 20 hotels (10 in Vienna, 10 in Munich) with attributes [4]:
 - *Price per night (USD), Distance from city center (km), Stars rating (1–5), Guest review rating (1–10), and Public transport accessibility score (1–10).*

¹<https://react.dev/>

²<https://vite.dev/>

³<https://katex.org/>

Dual-Algorithm System

The system implements two algorithmic modes that users can toggle [3]:

Algorithm A: Weighted MAUT

- **Approach:** Multi-Attribute Utility Theory (Additive) [5].
- **User Control:** Interactive sliders to define weights (w_i).
- **Output:** 0–100% match score and full ranking of all hotels.

Algorithm B: Constraint-Guided Filtering

- **Approach:** Hard filtering using user-defined bounds.
- **User Control:** Sliders for absolute limit thresholds.
- **Output:** Exact hotels satisfying all limits.

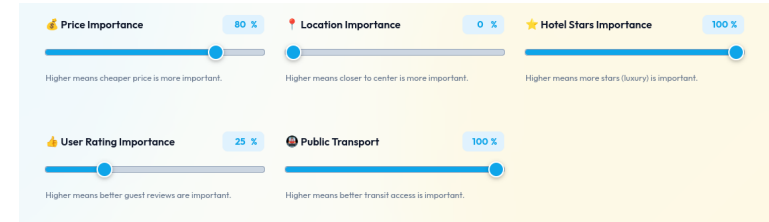


Abbildung: Implemented Sliders for Control

Deep Dive: Weighted MAUT Implementation

Calculates utility $U(X)$ for each hotel X [5]:

$$U(X) = \frac{\sum_{i=1}^5 w_i \cdot u_i(x_i)}{\sum_{i=1}^5 w_i} \times 100\%$$

Walkthrough:

- Attribute Utility ($u_i(x_i)$): hotel's raw values (like a price or distance) are normalized between 0 (worst option) and 1 (best option).
- Weighting (w_i): normalized value is multiplied by slider weight (from 0 to 100) and summed up for all weights and utilities.
- Division ($\sum w_i$): Finally, we divide by the sum of all weights.

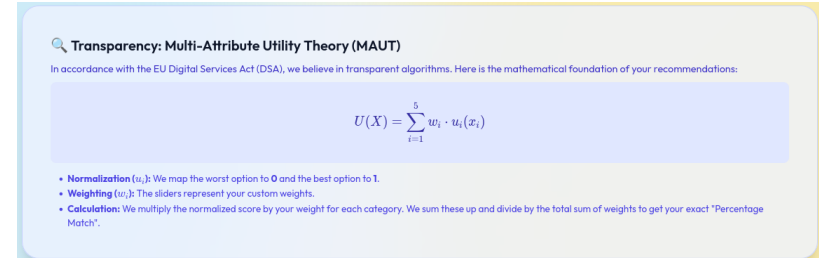


Abbildung: Transparency Banner in UI

Deep Dive: Constraint-Guided Filtering

A hard-limit filtering approach, showing exactly the fitting hotels

- **Pure Logic Over Complex Math:** Instead of calculating utility scores, system checks each hotel against user-defined limits.
- **Inequality Criteria Checked:**

Price $\leq$ Max Price

Distance $\leq$ Max Distance

Stars $\geq$ Min Stars

Rating $\geq$ Min Rating

Public Transport $\geq$ Min Transport

- **Highly Predictable Outcome:** Clear binary classification (a hotel is either included or excluded — no “black-box” scores).
- **Actionable Feedback:** When no hotels match the constraints, the system prompts user to ease their thresholds.

User Interface & DSA Compliance

Transparency is integrated into the user interface:

- **Global Explanation Banner:** Informative card at the top of the dashboard explaining the MAUT logic.
- **Interactive Calculations (Local Panel):**
 - Users can expand a “**Show Mathematical Breakdown**” panel on each hotel card.
 - Displays the step-by-step normalized values, weights, points, and final calculation.

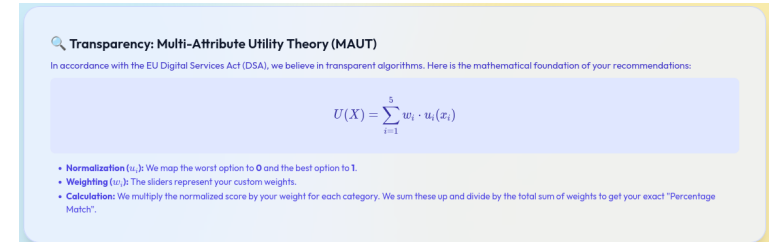


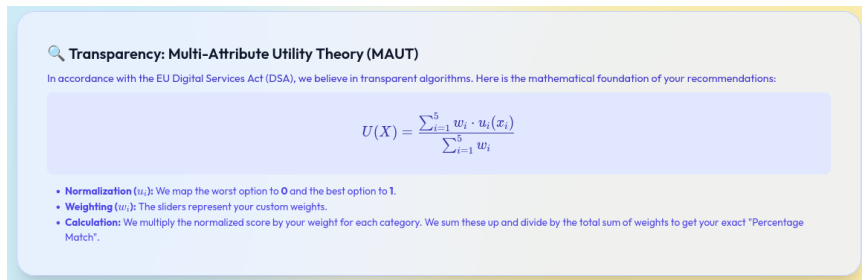
Abbildung: Global Transparency Banner in UI

Detailed Score Breakdown (Full Transparency)

To comply with DSA explainability standards, users can inspect the exact calculations on-demand:

Dynamic Calculation Steps

1. **Feature Normalization:** Converts raw hotel attributes (e.g. \$180 price) to a normalized value $u_i \in [0, 1]$ relative to the current city.
2. **Scaling by Weights:** Multiplies each u_i by the user's custom weight $w_i \in [0, 100]$.
3. **Aggregation:** Sums the individual weighted scores: $\sum w_i \cdot u_i(x_i)$.
4. **Final Division:** Divides the sum by the total active weights ($\sum w_i$) to yield the clean 0 – 100% match score.



Transparency: Multi-Attribute Utility Theory (MAUT)

In accordance with the EU Digital Services Act (DSA), we believe in transparent algorithms. Here is the mathematical foundation of your recommendations:

$$U(X) = \frac{\sum_{i=1}^5 w_i \cdot u_i(x_i)}{\sum_{i=1}^5 w_i}$$

- **Normalization** (u_i): We map the worst option to 0 and the best option to 1.
- **Weighting** (w_i): The sliders represent your custom weights.
- **Calculation:** We multiply the normalized score by your weight for each category. We sum these up and divide by the total sum of weights to get your exact "Percentage Match".

Abbildung: On-Demand Mathematical Breakdown Panel

Realized UI Dashboard

Key UI Elements:

- Clean, premium glassmorphism container design.
- Switchable cities: Munich and Vienna.
- Seamless sliders and dynamic input sync.
- Highlighted match badges with fluid transitions.
- Fully accessible and responsive across mobile and desktop.

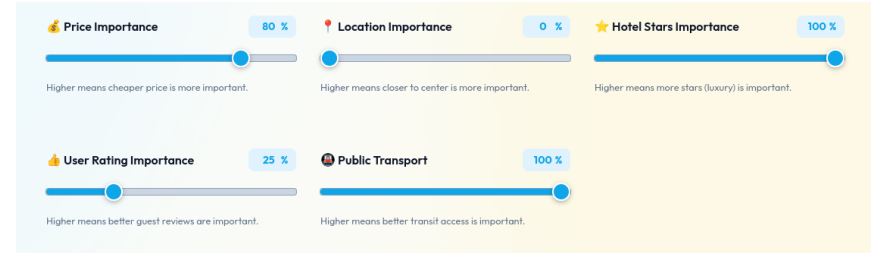


Abbildung: Implemented Web Application Interface

Evaluation & Future Directions

- **Summary of Achievements:**

- Successfully bridged the gap between complex recommendation logic and EU DSA transparency requirements.
- Implemented high-performance, real-time UI rendering of mathematical equations and data filtering.
- Received positive feedback on the click-to-expand mathematical explanation tool.

- **Future Work & Scaling:**

- **Hybrid Recommendation:** Combining filtering with MAUT (e.g. filter out > 300 and then weight remaining).
- **Critiquing Cycles:** Adding critique steps (“Find similar but cheaper”) using utility distance metrics.
- **Dynamic API Integration:** Fetching real-time hotel price and location data.

Thank you for your attention!

Questions?

References I

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- [2] European Parliament and Council of the European Union. *Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act)*. 2022. URL: <http://data.europa.eu/eli/reg/2022/2065/oj>.
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- [5] Christian Schmitt, Dietmar Dengler, Mathias Bauer u. a. „The maut-machine: an adaptive recommender system“. In: *Proceedings of the ABIS Workshop, Hannover, Germany*. 2002.